

HANFORD, CA, USA

WMO: 723898

Lat: 36.319N

Lon: 119.629W

Elev: 76

StdP: 100.42

Time Zone: -8.00 (NAP)

Period: 98-19

WBAN: 53119

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-2.0	-0.6	-5.3	2.4	8.5	-3.5	2.8	8.0	7.8	11.0	6.9	10.8	0.6	60	0.448

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.5	39.0	21.0	37.7	20.8	36.6	20.2	23.1	35.2	22.2	34.3	21.4	33.6	3.1	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	13.9	28.5	17.7	12.8	28.5	16.7	12.0	27.6	68.8	35.4	65.2	34.3	62.0	33.7	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.6	6.7	5.6	-3.9	41.5	1.6	1.3	-5.0	42.4	-6.0	43.2	-6.9	43.9	-8.0	44.8		
			-4.4	24.9	1.6	1.6	-5.6	26.1	-6.5	27.1	-7.4	28.0	-8.6	29.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.5	8.3	10.5	13.6	16.4	20.6	24.7	27.6	26.6	23.7	18.0	11.9	8.0
	DBStd	7.47	2.82	2.70	3.27	3.69	3.42	3.26	2.43	2.21	3.07	3.14	3.10	2.88
	HDD10.0	194	69	25	6	1	0	0	0	0	0	0	16	77
	HDD18.3	1336	313	221	149	78	16	2	0	0	2	42	193	321
	CDD10.0	2946	15	38	119	194	330	442	545	516	409	249	73	14
	CDD18.3	1045	0	0	3	22	87	193	287	258	162	33	1	0
	CDH23.3	15185	0	2	102	410	1278	2804	4164	3605	2224	568	28	1
	CDH26.7	8417	0	0	20	147	577	1581	2545	2150	1206	189	2	0
Wind	WSAvg	2.3	1.4	1.8	2.3	2.9	3.3	3.4	2.9	2.8	2.3	1.8	1.4	1.4
Precipitation	PrecAvg	218	42	39	39	18	6	2	0	0	5	11	24	29
	PrecMax	396	170	128	168	75	37	21	5	9	37	62	74	83
	PrecMin	93	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	77	35	35	34	19	9	4	1	2	9	14	20	22
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.7	23.3	29.0	33.7	37.2	40.1	41.1	40.2	38.5	34.0	26.3	19.0
		MCWB	11.8	14.5	16.4	17.8	20.7	21.5	21.6	21.0	20.1	19.0	15.7	13.0
	2%	DB	17.4	20.8	26.1	30.7	34.4	38.0	39.2	38.3	36.7	31.1	23.5	17.2
		MCWB	12.1	13.8	15.6	16.6	18.5	20.7	21.3	20.5	19.8	17.9	15.0	12.1
	5%	DB	15.6	18.5	23.7	27.9	32.2	36.3	37.9	37.2	35.2	28.7	21.2	15.3
		MCWB	11.5	12.8	14.8	15.7	17.7	19.7	21.3	20.4	19.2	16.6	14.0	11.1
	10%	DB	13.7	16.8	21.5	25.3	30.0	34.3	36.5	35.8	33.3	26.8	18.8	13.6
		MCWB	10.8	12.1	13.9	14.8	16.7	19.2	20.5	19.8	18.8	16.1	13.0	10.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.8	16.1	17.9	19.6	22.5	23.8	24.8	23.6	22.7	20.4	17.4	15.2
		MCDB	16.8	20.0	25.3	30.3	33.2	36.5	36.8	35.0	33.3	30.0	23.1	17.3
	2%	WB	13.2	14.7	16.7	17.7	20.1	22.4	23.4	22.4	21.1	19.0	15.9	13.4
		MCDB	15.5	19.0	24.2	28.0	32.0	34.9	35.6	34.9	33.3	27.9	21.6	15.4
	5%	WB	12.2	13.7	15.5	16.4	18.7	21.3	22.4	21.4	20.2	17.9	14.7	12.3
		MCDB	14.3	17.6	22.3	26.1	29.8	33.3	34.5	34.1	32.7	26.6	19.9	14.3
	10%	WB	11.0	12.5	14.4	15.3	17.4	20.1	21.5	20.6	19.3	16.8	13.6	10.9
		MCDB	13.4	16.0	20.3	24.1	28.4	32.0	33.7	33.3	31.8	25.0	17.9	12.8
Mean Daily Temperature Range	5% DB	MDBR	10.1	12.0	13.5	14.6	15.6	17.0	17.5	18.0	18.1	16.4	13.7	11.0
		MCDBR	13.7	15.6	17.4	19.1	19.1	19.4	18.7	19.4	20.4	19.7	17.3	13.6
		MCWBR	8.7	9.1	8.0	7.3	6.7	6.1	5.9	5.9	6.7	7.9	9.2	8.6
	5% WB	MCDBR	10.3	13.2	15.6	17.7	17.5	17.4	16.9	17.9	18.6	17.0	15.1	9.4
		MCWBR	7.0	8.0	7.7	7.2	6.6	6.1	6.2	6.3	6.6	7.2	8.5	6.5
Clear-Sky Solar Irradiance	taub	0.361	0.371	0.371	0.371	0.371	0.364	0.384	0.363	0.367	0.391	0.380	0.357	
	taud	2.313	2.304	2.320	2.346	2.367	2.386	2.358	2.393	2.392	2.272	2.252	2.329	
	Ebn at Noon	815	857	894	911	911	914	892	907	886	821	781	792	
	Edh at Noon	99	112	120	123	122	120	123	116	110	113	104	91	
All-Sky Solar Radiation	RadAvg	2.44	3.55	5.07	6.44	7.62	8.45	8.08	7.42	6.20	4.52	3.08	2.25	
	RadStd	0.35	0.39	0.42	0.45	0.39	0.25	0.28	0.20	0.17	0.23	0.17	0.30	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.80	N/A	N/A	+0.89	N/A	N/A	-92	-281	+318	+171
(47)	Regional (27 neighbors)	+0.57	N/A	-1.10	+0.55	N/A	N/A	-27	-148	+189	+61

Nomenclature: See separate page